



30 Minutes Expert

Recover deleted files

It's a habit that's hard to break. It seems so convenient, even secure, to permanently delete files. Surprisingly, most of us don't use the Recycle Bin's setting to permanently delete files. [Shift] + [Delete] is the vice. But moments later we realise that from the whole group, there was one file that is required. Or, a couple of days later, a client sends a mail saying that the project isn't complete and he'll need some files for cross-reference. Fortunately, there are ways to recover deleted files. All you need is the right tools. And a lot of luck.

Hard disk anatomy

A hard disk allows random data access, i.e. any file stored anywhere on it can be accessed at any time, as against a tape drive, where the tape would have to be wound to the particular location where the file is stored. Hard disks store data in sectors and clusters. The bytes of a file are allocated to each cluster, which are populated as required. Think of sectors and clusters as a hotel with floors and rooms where the files are guests. The guest list is stored at the reception and anyone behind the desk will be able to tell you which room a particular guest is in. Hard disks store information about each file in a similar way. The 'address' of each file is stored in a separate area of the hard disk that is not accessible with common tools. When a file is requested, the disk looks up this 'register', locates the file and passes it on. On FAT and FAT32 partitions, this entry is stored in the FAT (File Allocation Table).

When a file is permanently deleted, only its entry from the 'register' is removed. The actual file is still stored on the disk until some program requires

space. Thus, it may still be possible to recover the file until another overwrites it.

Stop!

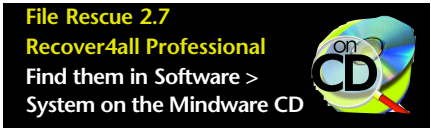
The moment you realise that you need to recover a deleted file, stop doing whatever you are doing on the PC. Do not close programs, shut down Windows, hit [Ctrl] + [S] or even chat, especially if you haven't defined a permanent swap file size in Windows. Any request for space, and the disk might just give up those precious clusters where the file may still be stored.

Plan out exactly how you want to go about recovering the file. Usually, there is no room for experimenting and, if any, there is only one chance. If the file is on a partition other than the Windows drive, you might have a much better chance of retrieving it.

Recover

It is advisable to have some recovery software already installed on the PC. Also make sure to install one on a drive other than the one running on Windows, as it may still write to the registry.

File Rescue: File Rescue (www.software-shelf.com) from Software Shelf is a small but useful tool for recovering files but the demo edition recovers only two files. It works with FAT, FAT32 and NTFS drives. When you launch File Rescue, it asks for the drives that are to be scanned and the kind of files to look for. Make your selections and click OK. It scans the disk quite fast and shows a list of files that it finds on the disk, but not in the FAT. It lists the file names, the folder from which

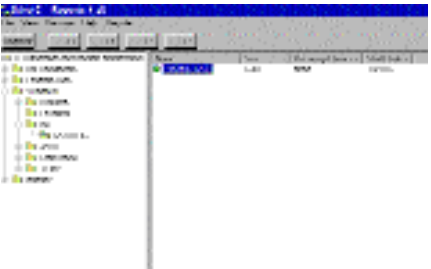


they were deleted, their sizes, modification dates, prognosis and the file address on the disk.

The most important thing here is the prognosis. If it says 'Poor', then some parts of the file have been overwritten by other data and the file probably won't recover correctly. If the condition is 'Good', then in most cases, the file will be recovered completely.

Select the files you want to recover and click Undelete. When recovering the files, choose a drive other than the one you are recovering from—you don't want any data to be written to the drive until you finish recovering.

Recover4all: Another utility, Recover4all (www.recover4all.com) is available in two editions—both support FAT and FAT32, but the professional version also supports NTFS. Launch the program and select the drive you want to scan for deleted files. It



Recover4all displays the deleted file in an Explorer-like view

Outlook Express Folders

If you have deleted a mail folder in Outlook Express, do not despair. Outlook Express stores folders as separate files. Look for files with your mailbox folder name and a DBX extension.

File Name	Folder	Size	Modified	Prognosis
1. Outlook Express Mail	C:\Program Files\Outlook Express\	1024	2001-01-01 10:00	Good
2. Outlook Express Mail	C:\Program Files\Outlook Express\	1024	2001-01-01 10:00	Good
3. Outlook Express Mail	C:\Program Files\Outlook Express\	1024	2001-01-01 10:00	Good
4. Outlook Express Mail	C:\Program Files\Outlook Express\	1024	2001-01-01 10:00	Good
5. Outlook Express Mail	C:\Program Files\Outlook Express\	1024	2001-01-01 10:00	Good
6. Outlook Express Mail	C:\Program Files\Outlook Express\	1024	2001-01-01 10:00	Good
7. Outlook Express Mail	C:\Program Files\Outlook Express\	1024	2001-01-01 10:00	Good
8. Outlook Express Mail	C:\Program Files\Outlook Express\	1024	2001-01-01 10:00	Good
9. Outlook Express Mail	C:\Program Files\Outlook Express\	1024	2001-01-01 10:00	Good
10. Outlook Express Mail	C:\Program Files\Outlook Express\	1024	2001-01-01 10:00	Good

File Rescue displays a list of deleted files with details on their condition

displays deleted files in an Explorer-like view, making it easier to navigate. This software also shows you the condition of the file and the possibility of recovering it correctly. Just select the files and click Recover.

Recovering files is simple and possible, but it doesn't always work. So the next time you use the [Shift] + [Delete] combination, be afraid. Be very afraid.

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